for some family of unitary operators $\hat{U}(t)$. If the Hamiltonian is independent of time then $\hat{U}(t) = e^{-i\hat{H}t}$ (in units where $\hbar = 1$).

Because of the unitarity of time evolution, all the basic operations in quantum computing and information processing (except measurements) are described by unitary operators.

2.1.3 Measurements and observables

The *measurement postulate* dictates how measurements on physical systems are modelled in quantum mechanics. We can describe measurements in two seemingly different but ultimately equivalent ways.

POVMs

A general measurement in quantum mechanics is associated with a set of measurement operators $\{\hat{M}_m\}$ where m labels the possible measurement outcomes. For a system in a normalised state $|\psi\rangle$, the probability for the outcome m to occur is

$$p(m) = \langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle. \tag{2.7}$$

This is known as the *Born rule*. Because this expression is the norm of the vector $\hat{M}_m|\psi\rangle$ (or equivalently, because the operator $\hat{E}_m = \hat{M}_m^\dagger \hat{M}_m$ is positive semidefinite), the probabilities are necessarily non-negative, $p(m) \geq 0$. In addition, the measurement operators must satisfy the completeness relation

$$\sum_m \hat{M}_m^\dagger \hat{M}_m = \mathbb{1}, \tag{2.8}$$

which ensures that the probabilities sum to one,

$$\sum_m p(m) = \sum_m \langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle = \langle\psi|(\sum_m \hat{M}_m^\dagger \hat{M}_m)|\psi\rangle = 1, \tag{2.9}$$

for any normalised state $|\psi\rangle$. The (normalised) state of the system immediately after a measurement with outcome m is given by

$$\frac{\hat{M}_m|\psi\rangle}{\sqrt{\langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle}} = \frac{1}{\sqrt{p(m)}}\hat{M}_m|\psi\rangle. \tag{2.10}$$

The set of positive operators $\{\hat{E}_m\}$ is called a *positive-operator-valued measure* (POVM). Any set of operators satisfying

$$\hat{E}_m^\dagger = \hat{E}_m \qquad \text{(Hermiticity)}, \tag{2.11}$$

$$\hat{E}_m \geq 0 \qquad \text{(positivity)}, \tag{2.12}$$

$$\sum_m \hat{E}_m = \mathbb{1} \qquad \text{(completeness)}, \tag{2.13}$$

form a POVM and the operators $\hat{E}_m$ are referred to as POVM elements. POVMs provide the most general description of measurements in quantum mechanics. The outcome probabilities are given by $p(m) = \langle\psi|\hat{E}_m|\psi\rangle$. Note that the decomposition of a POVM into measurement operators, $\hat{E}_m = \hat{M}_m^\dagger \hat{M}_m$,